Do Now · retrieval

Lesson 100 · GPUs: Parallel Work Beyond Graphics

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which processor is designed for throughput on massive parallel workloads?

Q2

What is parallelism?

Q3

What is a data dependency?

Q4

An image array has 4 rows × 3 columns. How many pixels?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — GPUs provide many execution units designed for high-throughput parallel operations on large batches of data.

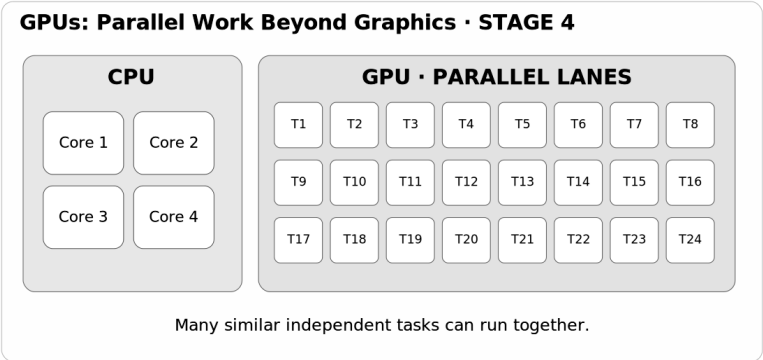
Core idea 2

Apply — Besides graphics, GPUs can accelerate machine learning, physical simulations, image processing and some scientific calculations.

Core idea 3

Evaluate — Transfers between memory/device, limited parallelism and divergent branching can outweigh GPU speedup on small or sequential tasks.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

GPU — Graphics Processing Unit.

SIMT — Single instruction, multiple threads execution model.

throughput — Total work per time.

divergence — Threads follow differing control paths.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

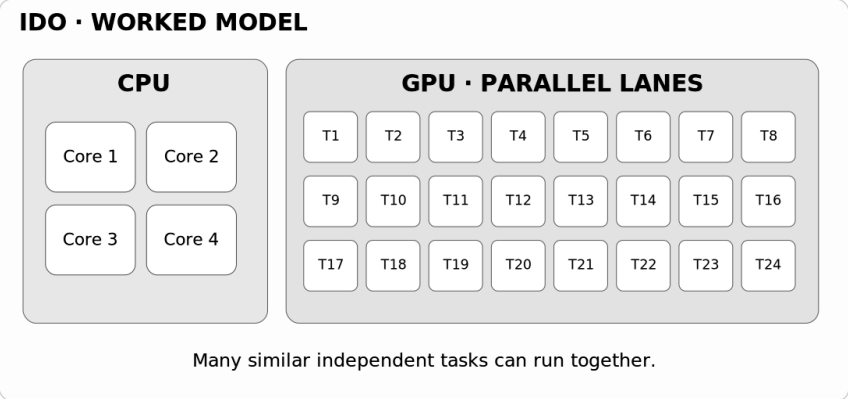
Problem

Explain why processing one million independent pixels may suit a GPU.

Follow the worked method

1. Follow the reasoning — Each pixel can receive similar operation in parallel; many GPU threads process different positions.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
Not necessarily: dependence limits simultaneous work, and data-transfer overhead may dominate.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain two features that make a GPU useful for applying same filter to millions of pixels. [6]

Question 2

Give two non-graphics GPU applications and explain how parallelism helps. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare CPU and GPU for 12-item sequential task versus million-element independent matrix operations. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A GPU is used exclusively for drawing computer-game graphics.

Use key vocabulary from memory

1. Explain GPU in your own words.

2. Explain SIMT in your own words.

3. Explain throughput in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.